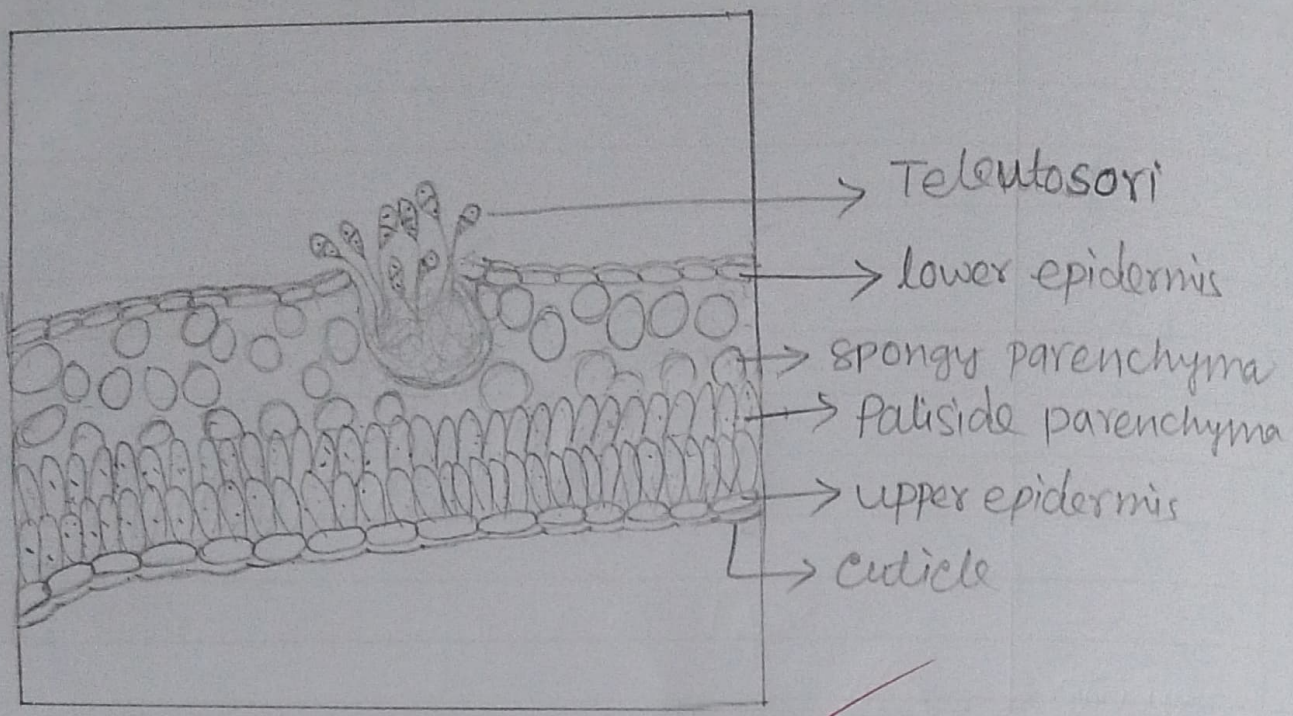
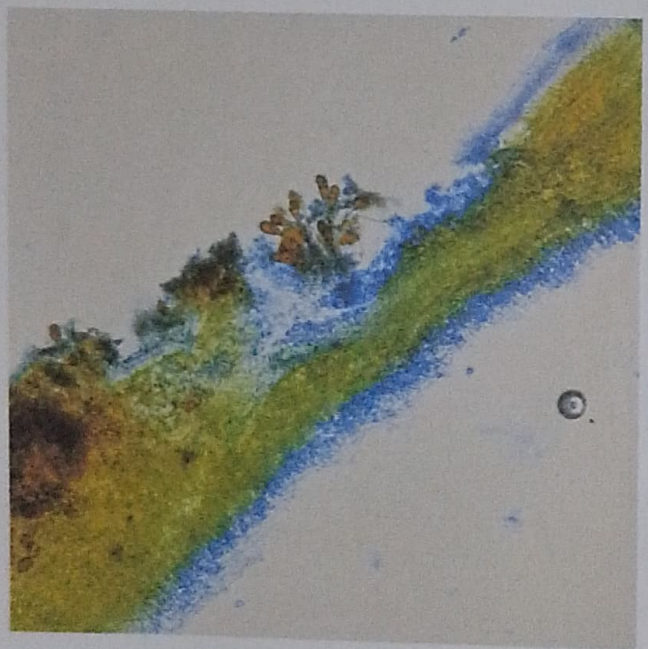


Fig- T.S. of infected leaf



Date 07/08/2026

Expt. No. _____ Page No. 23

Study of Rust of Justicia.

ø Symptoms-

i) Characteristic reddish-brown to dark-brown rust postulets are present on the leaf surface, petiole and stem.

ii) The postules are small, round to elongated in shape and appear as rusty spots.

iii) In severe infection, postules may coalesce with one another to form larger infected patches.

ø Histopathology -

i) A section through the infected region reveals intercellular fungal mycelium and sori containing numerous teliospores.

ii) The fungal mycelium grows within the host tissues and produces spore-bearing structure beneath the epidermis.

iii) The host epidermis ruptures due to the pressure exerted by the developing sori.

iv) Teliospores are produced terminally on the fungal hyphae

v) Each cell of the teliospore contains a single nucleus. (2n).

vi) The apex of the teliospore may be rounded or slightly pointed.

Teacher's Signature _____

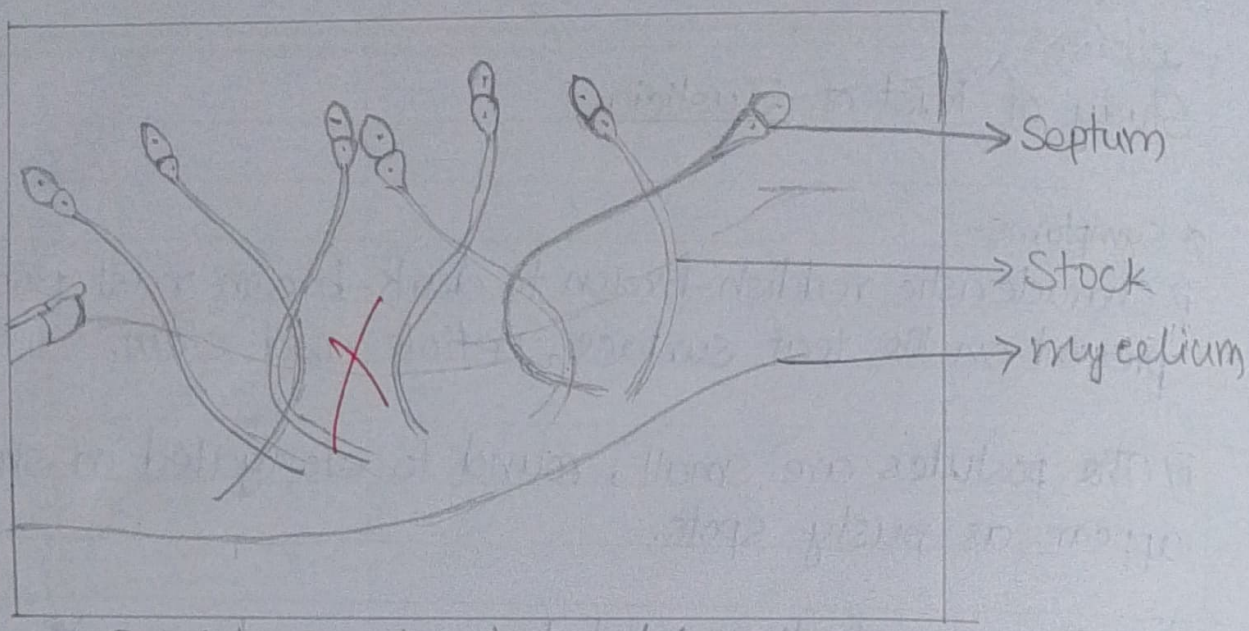


Fig- Microscopic view at 40x

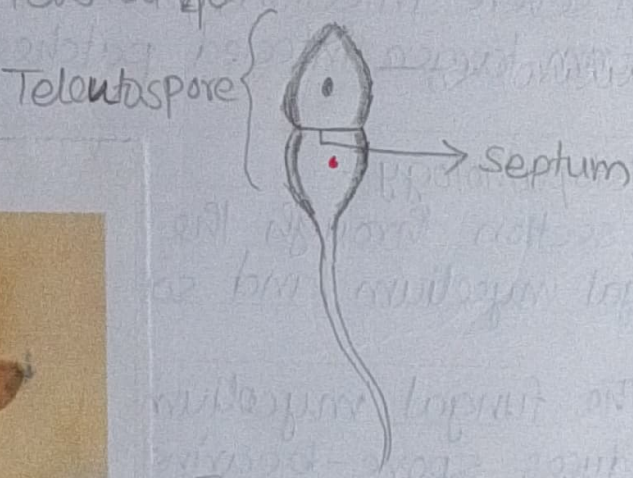


Fig- Single teleutospore



Date 07/08/2026

Expt. No. _____ Page No. 24

Ø Comments –

i) Reddish-brown rust pustules are present on the infected leaves and stems.

ii) Groups of teliospores are present in the sori.

iii) Teliospores are brown, stalked, two-celled and thick walled.

iv) The fungal infection causes rust-like symptoms on the host plant.

v) The specimen is identified as Rust of Justicia caused by the pathogenic fungus Puccinia ~~Justiciae~~ bagyanarayanii.

26/09/26

Teacher's Signature _____